

# Finding 2nd order stationary markov processes

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## Definitions

A stochastic process  $X(t)$  is second-order stationary if:

1.  $E[X(t)] = \mu$  (a constant).
2.  $\text{Cov}(X(s), X(t)) = r_{X(|t-s|)}$ .

## The Inciting Idea

In chapter 4.2 of Hoel Port and Stone, there is a proof showing that if a 2 state birth death chain starts at its steady state, it is 2nd order stationary.

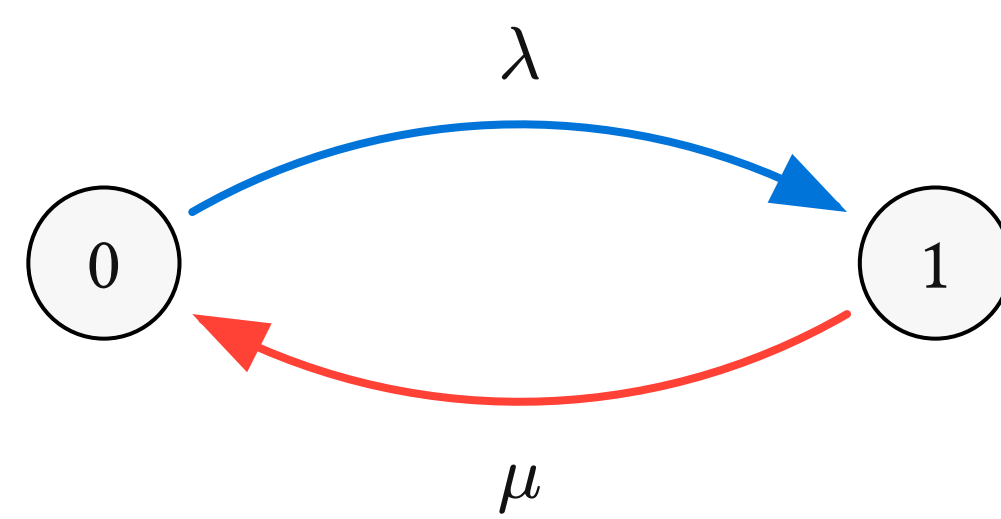


Figure 1: A confirmed 2nd order stationary process

The natural question is then ask if this is true for a 3 node birth-death process, or any 3 node chain in ?

## Proof for 3 Node Birth-Death Chain

We will informally show that a 3 node birth death chain is second order stationary, while the general case can just as easily have been shown here, I opted for a numerical example to hopefully be more easy to follow for the reader. Take the following Markov process.

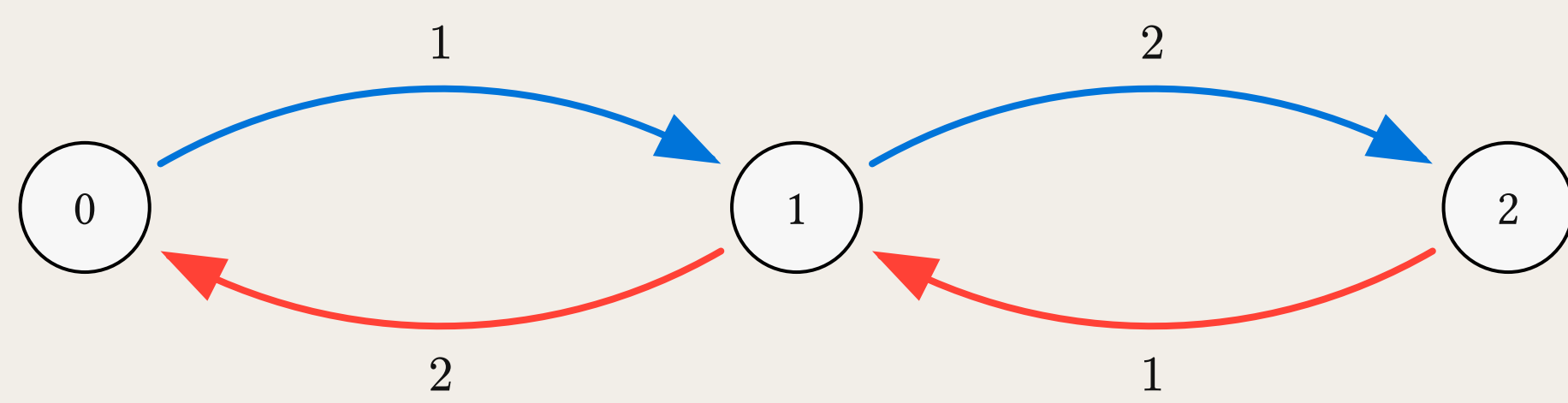


Figure 2: A birth Death Process :)

The steady state of this birth death chain is  $\vec{\pi} = [\frac{2}{5}, \frac{1}{5}, \frac{2}{5}]$

The p-matrix for this process is:

$$\frac{e^{-5t}}{10} \begin{pmatrix} 5e^{4t} + 4e^{5t} + 1 & -2(1 - e^{5t}) & -5e^{4t} + 4e^{5t} + 1 \\ -4(1 - e^{5t}) & 2(e^{5t} + 4) & -4(1 - e^{5t}) \\ -5e^{4t} + 4e^{5t} + 1 & -2(1 - e^{5t}) & 5e^{4t} + 4e^{5t} + 1 \end{pmatrix}$$

Given that the initial state of the Markov process is at its steady state we will show that  $E[X(t)] = \mu$  for all of t.

To prove the mean is constant, we first need to find the probability distribution of  $X(t)$  for all of t, of each state.

We can use the Total Law of Probability to find probability at state 0

$$P(X(t) = 0) = P(X(0) = 0, X(t) = 0) + P(X(0) = 1, X(t) = 0) + P(X(0) = 2, X(t) = 0)$$

Note that  $P(a, b) = P(a) \cdot P(b|a)$

$$= \frac{2}{5}P_{0,0}(t) + \frac{1}{5}P_{0,1}(t) + \frac{2}{5}P_{0,2}(t) = \frac{4}{25} + \frac{2}{25} + \frac{4}{25} = \frac{2}{5}$$

(I know I am skipping steps but the equation went off the screen)

Doing the same thing for state 1 and state 2 we have

$$P(X(t) = 2) = \frac{2}{5}; P(X(t) = 1) = \frac{1}{5}$$

Thus

$$E(x(t)) = 0 * \frac{2}{5} + 1 * \frac{1}{5} + 2 * \frac{2}{5} = 1$$

which is a constant

Now we will show that the Covariance is determined only by difference in time of the elements we are comparing. (This is the hard part)

Without loss of generality assume  $s < t$  then  $\text{Cov}(X(s), X(t))$

$$= E[X(s) \cdot X(t)] - E[X(s)] \cdot E[X(t)] = E[X(s) \cdot X(t)] - 1 \cdot 1$$

Consider the joint distribution table of  $[X(s), X(t)]$

| $X(s) \setminus X(t)$ | 0 | 1                       | 2                       |
|-----------------------|---|-------------------------|-------------------------|
| 0                     | — | —                       | —                       |
| 1                     | — | $P[X(s) = 1, X(t) = 1]$ | $P[X(s) = 2, X(t) = 1]$ |
| 2                     | — | $P[X(s) = 1, X(t) = 2]$ | $P[X(s) = 2, X(t) = 2]$ |

$$\text{Cov}(X(s), X(t)) = P[X(s) = 1, X(t) = 1] + 2P[X(s) = 2, X(t) = 1] + 2P[X(s) = 1, X(t) = 2] + 4P[X(s) = 2, X(t) = 2]$$

Simplifying we get

$$\text{Cov}(X(s), X(t)) = \frac{1}{5}(P_{1,1})(t-s) + \frac{4}{5}(P_{2,1}(t-s)) + \frac{2}{5}(P_{1,2}(t-s)) + \frac{8}{5}(P_{2,2}(t-s)) - 1$$

Note that the Covariance is a composition of (t-s) as desired

We have shown that the Markov process is second order stationary.  $\square$

## What about other 3 node Markov processes?

Looking into the following Markov process, we can easily show that given it starting at its steady state that its expected value at any point of in time will be its steady state (as that is the definition of a steady state)

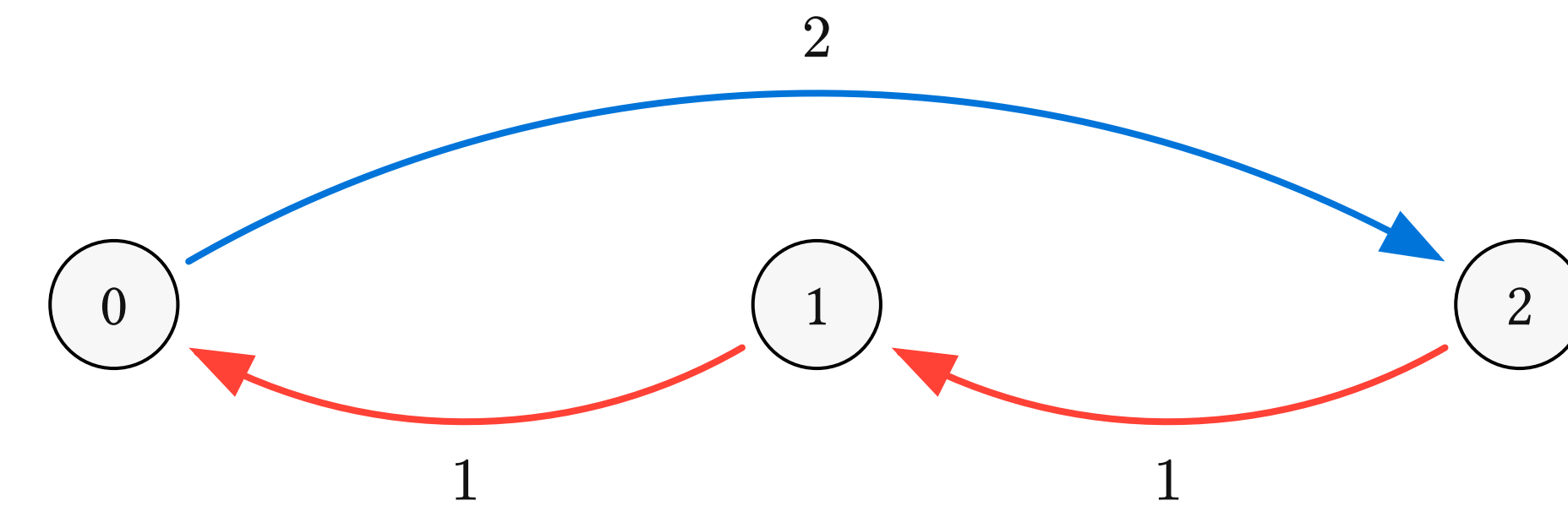


Figure 3: A pretty P matrix generating Markov process

The P Matrix is:

$$\frac{e^{-2t}}{5} \begin{pmatrix} 2e^{2t} + \sin(t) + 3 \cos(t) & e^{2t} - 2 \sin(t) - \cos(t) & 2e^{2t} + \sin(t) - 2 \cos(t) \\ -2(-e^{2t} - 3 \sin(t) + \cos(t)) & e^{2t} - 2 \sin(t) + 4 \cos(t) & -2(-e^{2t} + 2 \sin(t) + \cos(t)) \\ -2(-e^{2t} + 2 \sin(t) + \cos(t)) & e^{2t} + 3 \sin(t) - \cos(t) & 2e^{2t} + \sin(t) + 3 \cos(t) \end{pmatrix}$$

The steady state of this birth death chain is  $\vec{\pi} = [\frac{2}{5}, \frac{1}{5}, \frac{2}{5}]$  (yes I did pick a Markov chain that has the same steady state)

Thus we will jump to showing that the Covariance satisfies the conditions of a second order stationary process

Without loss of generality assume  $s < t$  then  $\text{cov}(X(s), X(t))$

$$= E[X(s) \cdot X(t)] - E[X(s)] \cdot E[X(t)] = E[X(s) \cdot X(t)] - 1$$

Consider the joint distribution table of  $[X(s), X(t)]$

| $X(s) \setminus X(t)$ | 0 | 1                       | 2                       |
|-----------------------|---|-------------------------|-------------------------|
| 0                     | — | —                       | —                       |
| 1                     | — | $P[X(s) = 1, X(t) = 1]$ | $P[X(s) = 1, X(t) = 2]$ |
| 2                     | — | $P[X(s) = 2, X(t) = 1]$ | $P[X(s) = 2, X(t) = 2]$ |

$$\text{cov}(X(s), X(t)) = P[X(s) = 1, X(t) = 1] + 2P[X(s) = 2, X(t) = 1] + 2P[X(s) = 1, X(t) = 2] + 4P[X(s) = 2, X(t) = 2] - 1$$

Simplifying we get

$$\text{cov}(X(s), X(t)) = \frac{1}{5}(P_{1,1})(t-s) + \frac{4}{5}(P_{2,1}(t-s)) + \frac{2}{5}(P_{1,2}(t-s)) + \frac{8}{5}(P_{2,2}(t-s)) - 1$$

Note that the covariance is a composition of (t-s) as desired

We have shown that the Markov process is second order stationary.  $\square$

## Further Steps and Exploration into Generalization

While I could love to make the statement that any Markov process is second-order stationary **if and only if** it is initialized in its stationary distribution  $\pi$ . There is far more work to be done to assume this is a correct assertion.

**Potential Avenues of Exploration** The goal is to try and identify generalizing patterns in the approaches in finding Markov processes as second order stationary

- Examining different structures of Markov chains
- Expanding to N-size Markov chains
- Exploration into non-homogenous Markov chains with a steady state

## References

- [1] P. G. Hoel, S. C. Port, and C. J. Stone, *Introduction to Stochastic Processes*. Waveland Press, 1987.

## Acknowledgements

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